

# IIE7585 Midterm Exam

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## Problem 1

### Sets/Indices

- $p \in P$ : product set = {desk, table, chair}
- $r \in R$ : resource set = {lumber, finishing, carpentry}
- $s \in S$ : scenario set = {low, medium, high}
- $\omega \in \Omega$ : set of all possible scenarios

### Parameters

- $a_{rp}$ : amount of resource  $r$  needed to produce one unit of product  $p$

$$A = [a_{rp}] = \begin{bmatrix} 8 & 6 & 1 \\ 4 & 2 & 1.5 \\ 2 & 1.5 & 0.5 \end{bmatrix}$$

- $b_r$ : total availability of resource  $r$

$$b = [b_r] = \begin{bmatrix} 3500 \\ 1500 \\ 1000 \end{bmatrix}$$

- $c_r$ : unit cost of purchasing one unit of resource  $r$

$$c = [c_r] = \begin{bmatrix} 2 \\ 4 \\ 5.2 \end{bmatrix}$$

- $\rho_p$ : selling price per unit of product  $p$

$$\rho = [\rho_p] = \begin{bmatrix} 60 \\ 40 \\ 10 \end{bmatrix}$$

- $d_p(\omega^s)$ : demand for product  $p$  in scenario  $s$

$$d(\omega) = [d_{ps}] = \begin{bmatrix} 50 & 150 & 250 \\ 20 & 110 & 250 \\ 200 & 225 & 500 \end{bmatrix}$$

- $\pi(\omega^s)$ : probability of scenario  $s$

$$\pi(\omega) = [\pi_s] = \begin{bmatrix} 0.3 \\ 0.4 \\ 0.3 \end{bmatrix}$$

## Decision Variables

- First-stage:

◦  $x_r$ : amount of resource  $r$  to be purchased

$$x = [x_r] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- Second-stage:

◦  $y_p(\omega^s)$ : amount of product  $p$  to be produced in scenario  $s$

$$y(\omega) = [y_{ps}] = \begin{bmatrix} y_{11} & y_{12} & y_{13} \\ y_{21} & y_{22} & y_{23} \\ y_{31} & y_{32} & y_{33} \end{bmatrix}$$

### (a) DEP

#### DEP of the two-stage formulation

$$\max \quad -c^\top x + \sum_{\omega \in \Omega} \pi(\omega) \rho^\top y(\omega)$$

$$\text{s.t.} \quad x \leq b$$

$$-x + Ay(\omega) \leq 0$$

$$y(\omega) \leq d(\omega)$$

$$x, y(\omega) \geq 0$$

$$\forall \omega \in \Omega$$

$$\forall \omega \in \Omega$$

$$\forall \omega \in \Omega$$

#### Dual formulation to the DEP

$$\min \quad b^\top \lambda + \sum_{\omega \in \Omega} d(\omega)^\top \nu(\omega)$$

$$\text{s.t.} \quad \lambda - \sum_{\omega \in \Omega} \mu(\omega) \geq -c$$

$$A^\top \mu(\omega) + \nu(\omega) \geq \pi(\omega) \rho$$

$$\lambda, \mu(\omega), \nu(\omega) \geq 0$$

$$(1a)$$

$$(1b)$$

$$(1c)$$

$$(1d)$$

$$(1e)$$

$$(2a)$$

$$(2b)$$

$$(2c)$$

$$(2d)$$

#### Comparison between DEP and Dual Formulation

- The DEP is a maximization problem aiming to maximize profit, given by the total expected revenue minus the total cost. The dual is a minimization problem that seeks to minimize the cost of resource usage through shadow prices and the cost of unmet demand.
- In both formulations, the second-stage variables ( $y(\omega)$  in the DEP,  $\mu(\omega)$  and  $\nu(\omega)$  in the dual) are scenario-specific. However, the linking variable ( $x$  in the DEP,  $\lambda$  in the dual) couples the scenarios together.
- The first-stage decision variable  $x$  serves as the linking variable in the DEP, equal across all scenarios. In the dual, the variable  $\lambda$  links the first-stage constraints across all scenarios.
- In the DEP, the constraint (1c) couples  $x$  with each scenario's production decisions. In the dual, the constraint (2b) ensures that the sum of scenario-specific dual variables  $\mu(\omega)$  collectively balance the unit cost coefficients.

### (b) DEP-S

To write the scenario formulation, let  $x(\omega)$  be copies of the first-stage decision variable  $x$  for each  $\omega \in \Omega$ . The expectation nonanticipativity constraint (3e) is added to guarantee that  $x(\omega)$  is the same for all  $\omega \in \Omega$ .

$$\max \quad \sum_{\omega \in \Omega} \pi(\omega) (-c^\top x(\omega) + \rho^\top y(\omega)) \quad (3a)$$

$$\text{s.t.} \quad x(\omega) \leq b$$

$$-x(\omega) + Ay(\omega) \leq 0$$

$$y(\omega) \leq d(\omega)$$

$$x(\omega) = \sum_{\bar{\omega} \in \Omega} \pi(\bar{\omega}) x(\bar{\omega})$$

$$x(\omega), y(\omega) \geq 0$$

$$\forall \omega \in \Omega$$

$$\forall \omega \in \Omega$$

$$\forall \omega \in \Omega$$

$$\forall \omega \in \Omega$$

$$\forall \omega \in \Omega$$

$$(3b)$$

$$(3c)$$

$$(3d)$$

$$(3e)$$

$$(3f)$$

### (c) Stage-wise (Nested) Multistage Formulation

Let  $\Omega_1$  and  $\Omega_2$  denote the set of second-stage and third-stage scenarios, respectively, with  $\omega_1 \in \Omega_1$  and  $\omega_2 \in \Omega_2$ . Let  $\pi(\omega_1)$  and  $\pi(\omega_2)$  be their respective probabilities.

$$d(\omega_1) = \begin{bmatrix} 50 & 150 & 250 \\ 20 & 110 & 250 \\ 200 & 225 & 500 \end{bmatrix} \quad d(\omega_2) = \begin{bmatrix} 50 & 100 & 200 \\ 50 & 120 & 200 \\ 150 & 200 & 300 \end{bmatrix}$$

$$\pi(\omega_1) = [0.3 \quad 0.4 \quad 0.3]^\top \quad \pi(\omega_2) = [0.4 \quad 0.2 \quad 0.4]^\top$$

Let  $x_r^1$  denote the amount of resource  $r$  purchased at stage 1 (here-and-now). Let  $x_r^2(\omega_1)$  be the amount of resource  $r$  purchased at stage 2, after the realization of scenario  $\omega_1$ . Similarly,  $x_r^3(\omega_1, \omega_2)$  represents the amount of resource  $r$  purchased at stage 3, given the scenario path  $(\omega_1, \omega_2)$ . Let  $y_p^2(\omega_1)$  denote the amount of product  $p$  produced at stage 2 under scenario  $\omega_1$ , and let  $y_p^3(\omega_1, \omega_2)$  denote the amount of product  $p$  produced at stage 3 under scenario path  $(\omega_1, \omega_2)$ .

The stage-wise (nested) multistage formulation of this problem is the following:

$$\max \quad -c^\top x^1 + \sum_{\omega_1 \in \Omega_1} \pi(\omega_1) \left[ \rho^\top y^2(\omega_1) - c^\top x^2(\omega_1) \right] \quad (4a)$$

$$+ \sum_{\omega_2 \in \Omega_2} \pi(\omega_2) \left( \rho^\top y^3(\omega_1, \omega_2) - c^\top x^3(\omega_1, \omega_2) \right) \Big]$$

$$\text{s.t.} \quad x^1 \leq b \quad (4b)$$

$$Ay^2(\omega_1) \leq x^1 \quad \forall \omega_1 \in \Omega_1 \quad (4c)$$

$$y^2(\omega_1) \leq d(\omega_1) \quad \forall \omega_1 \in \Omega_1 \quad (4d)$$

$$x^2(\omega_1) \leq b \quad \forall \omega_1 \in \Omega_1 \quad (4e)$$

$$Ay^3(\omega_1, \omega_2) \leq x^1 + x^2(\omega_1) - Ay^2(\omega_1) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (4f)$$

$$y^3(\omega_1, \omega_2) \leq d(\omega_2) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (4g)$$

$$x^3(\omega_1, \omega_2) \leq b \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (4h)$$

$$x^1, y^2(\omega_1), x^2(\omega_1), y^3(\omega_1, \omega_2), x^3(\omega_1, \omega_2) \geq 0 \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (4i)$$

### (d) Scenario Formulation of the Multistage Problem

Let  $\Omega_1$  and  $\Omega_2$  be the sets of scenarios for the second and third stages, respectively. To write the scenario formulation, let  $x^1(\omega_1, \omega_2)$ ,  $x^2(\omega_1, \omega_2)$ , and  $x^3(\omega_1, \omega_2)$  be copies of the first-, second-, and third-stage decision variables for each full scenario  $(\omega_1, \omega_2) \in \Omega_1 \times \Omega_2$ . The nonanticipativity constraints (5i) and (5j) are added to ensure that  $x^1(\omega_1, \omega_2)$  is the same across all scenarios, and that  $x^2(\omega_1, \omega_2)$  is the same across scenarios sharing the same  $\omega_1$ .

$$\max \quad \sum_{\omega_1 \in \Omega_1} \sum_{\omega_2 \in \Omega_2} \pi(\omega_1) \pi(\omega_2) \left( -c^\top x^1(\omega_1, \omega_2) + \rho^\top y^2(\omega_1, \omega_2) \right) \quad (5a)$$

$$- c^\top x^2(\omega_1, \omega_2) + \rho^\top y^3(\omega_1, \omega_2) - c^\top x^3(\omega_1, \omega_2) \Big)$$

$$\text{s.t.} \quad x^1(\omega_1, \omega_2) \leq b \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5b)$$

$$Ay^2(\omega_1, \omega_2) \leq x^1(\omega_1, \omega_2) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5c)$$

$$y^2(\omega_1, \omega_2) \leq d(\omega_1) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5d)$$

$$x^2(\omega_1, \omega_2) \leq b \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5e)$$

$$Ay^3(\omega_1, \omega_2) \leq x^1(\omega_1, \omega_2) + x^2(\omega_1, \omega_2) - Ay^2(\omega_1, \omega_2) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5f)$$

$$y^3(\omega_1, \omega_2) \leq d(\omega_2) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5g)$$

$$x^3(\omega_1, \omega_2) \leq b \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5h)$$

$$x^1(\omega_1, \omega_2) = \sum_{\bar{\omega}_1 \in \Omega_1} \sum_{\bar{\omega}_2 \in \Omega_2} \pi(\bar{\omega}_1) \pi(\bar{\omega}_2) x^1(\bar{\omega}_1, \bar{\omega}_2) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5i)$$

$$x^2(\omega_1, \omega_2) = \sum_{\bar{\omega}_2 \in \Omega_2} \pi(\bar{\omega}_2) x^2(\bar{\omega}_1, \bar{\omega}_2) \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5j)$$

$$x^1(\omega_1, \omega_2), x^2(\omega_1, \omega_2), x^3(\omega_1, \omega_2), y^2(\omega_1, \omega_2), y^3(\omega_1, \omega_2) \geq 0 \quad \forall \omega_1 \in \Omega_1, \omega_2 \in \Omega_2 \quad (5k)$$

The stage-wise (nested) formulation models the sequential decision-making process more naturally by moving across stages throughout a clear time flow, whereas the scenario formulation replicates decision variables across full scenario paths and requires explicit nonanticipativity constraints to ensure consistency. While the stage-wise model is compact and intuitive, the scenario model is more general and suitable for complex scenario trees at the cost of larger problem size.

## Problem 2

### (a)

We aim to prove that the recourse function  $f(x, \omega)$  is a convex function in  $(r(\omega), T(\omega))$ .

The recourse function is defined in this way:  $f(x, \omega) = \min \{q(\omega)^\top y \mid Wy \geq r(\omega) - T(\omega)x, y \geq 0\}$ .

The dual of this primal problem is:  $\max \{\mu^\top (r(\omega) - T(\omega)x) \mid W^\top \mu \leq q(\omega), \mu \geq 0\}$ .

By strong duality, the optimal value satisfies:  $f(x, \omega) = \mu^\top (r(\omega) - T(\omega)x)$  where  $\mu$  is an optimal dual solution.

Having  $x$  as a constant, consider the convex combination  $(r_\lambda, T_\lambda) = \lambda(r_1, T_1) + (1 - \lambda)(r_2, T_2)$  for any  $\lambda \in [0, 1]$ .

We want to show that  $f(x; r_\lambda, T_\lambda) \leq \lambda f(x; r_1, T_1) + (1 - \lambda)f(x; r_2, T_2)$  to prove convexity.

Setting  $\mu_1$  and  $\mu_2$  as optimal dual solutions corresponding to  $(r_1, T_1)$  and  $(r_2, T_2)$ ,

$f(x; r_1, T_1) = \mu_1^\top (r_1 - T_1 x)$ ,  $f(x; r_2, T_2) = \mu_2^\top (r_2 - T_2 x)$  holds owing to strong duality.

Using the definition of dual feasible solutions,  $\mu_1$  and  $\mu_2$  are also feasible for the dual of the convex combination problem, because the constraints are linear.

Thus, evaluating at  $\mu_1$  and  $\mu_2$  for the convex combination  $(r_\lambda, T_\lambda)$ ,

$$\begin{aligned} f(x; r_\lambda, T_\lambda) &= \min \mu^\top (r_\lambda - T_\lambda x) \\ &\leq \lambda \mu_1^\top (r_1 - T_1 x) + (1 - \lambda) \mu_2^\top (r_2 - T_2 x) \\ &= \lambda f(x; r_1, T_1) + (1 - \lambda) f(x; r_2, T_2). \end{aligned}$$

Therefore, by the definition of convexity,  $f(x, \omega)$  is convex in  $(r(\omega), T(\omega))$ . ■

### (b)

We aim to prove that the recourse function  $f(x, \omega)$  is a convex function in  $x$  for all  $x \in X_1 \cap X_2$ .

Recall that the recourse function is defined as:  $f(x, \omega) = \min \{q(\omega)^\top y \mid Wy \geq r(\omega) - T(\omega)x, y \geq 0\}$ .

Having  $\lambda \in [0, 1]$ , we want to show  $f(\lambda x_1 + (1 - \lambda)x_2, \omega) \leq \lambda f(x_1, \omega) + (1 - \lambda)f(x_2, \omega)$  for any  $x_1, x_2 \in X_1 \cap X_2$  and fixed  $\omega \in \Omega$ .

Let  $y_1$  and  $y_2$  be optimal solutions corresponding to  $x_1$  and  $x_2$ , respectively, such that:

$$\begin{aligned} y_1 &\in \arg \min \{q(\omega)^\top y \mid Wy \geq r(\omega) - T(\omega)x_1, y \geq 0\}, \\ y_2 &\in \arg \min \{q(\omega)^\top y \mid Wy \geq r(\omega) - T(\omega)x_2, y \geq 0\}. \end{aligned}$$

Now, define  $y_\lambda = \lambda y_1 + (1 - \lambda)y_2$ .

Since  $Wy_1 \geq r(\omega) - T(\omega)x_1$  and  $Wy_2 \geq r(\omega) - T(\omega)x_2$ , we have:

$$\begin{aligned} Wy_\lambda &= \lambda Wy_1 + (1 - \lambda)Wy_2 \geq \lambda(r(\omega) - T(\omega)x_1) + (1 - \lambda)(r(\omega) - T(\omega)x_2). \\ Wy_\lambda &\geq r(\omega) - T(\omega)(\lambda x_1 + (1 - \lambda)x_2). \end{aligned}$$

Also,  $y_\lambda \geq 0$  since it is a convex combination of nonnegative vectors. Therefore,  $y_\lambda$  is a feasible solution for  $f(\lambda x_1 + (1 - \lambda)x_2, \omega)$ .

Evaluating the objective at  $y_\lambda$ ,

$$\begin{aligned} q(\omega)^\top y_\lambda &= \lambda q(\omega)^\top y_1 + (1 - \lambda)q(\omega)^\top y_2 \\ &= \lambda f(x_1, \omega) + (1 - \lambda)f(x_2, \omega). \end{aligned}$$

Since  $f(\lambda x_1 + (1 - \lambda)x_2, \omega)$  is the minimum over all feasible  $y$ , we conclude

$$f(\lambda x_1 + (1 - \lambda)x_2, \omega) \leq \lambda f(x_1, \omega) + (1 - \lambda)f(x_2, \omega).$$

Hence,  $f(x, \omega)$  is convex in  $x$  over  $X_1 \cap X_2$ . ■